

TECHNICAL QUALIFICATIONS SUPPLEMENT

1. FORMAL EDUCATION

Hennepin Technical College | Four Associate of Applied Science Degrees

- Fluid Power Engineering Technician — AAS
- Hydraulic Engineering Technician — AAS
- Pneumatic Engineering Technician — AAS
- Automation Robotics Engineering Technology — AAS

Additional education: Emergency medical technician training (2000–2001).

2. PROFESSIONAL CERTIFICATIONS & VENDOR TRAINING

IFPS: Fluid Power Electronic Controls Specialist (CFPS); Pneumatic Specialist (PS); Hydraulic Specialist (HS)

FANUC: HandlingTool Operations & Programming J2P0310; iRVision Operations & Programming 2D

PMMI: Industrial Electricity 1; Fluid Power 1; PLC 1; Mechatronics Mechanical Components 1

SOLIDWORKS: SOLIDWORKS Electrical - Schematic

IPC: IPC-A-610 acceptability of electronic assemblies

3. SELECTED COLLEGE TECHNICAL COURSEWORK

Fluid Power: Fluid Power Technology I; Fluid Power Technology II; Fluid Power Motion Control; Hydraulic Specialist Certification Review; Pneumatic Specialist Certification Review.

Electrical / Automation / Controls: Industrial Electricity and Electronics I & II; Programmable Logic Controllers; Advanced Programmable Logic Controllers; Automation Controls; Advanced Automation Controls.

Instrumentation / Process Control: Instrumentation and Process Control I; Instrumentation and Process Control II.

Industrial Machinery: Packaging Machinery Systems; Troubleshooting Packaging Machinery.

Networking / Industrial Communications: communication models, network protocols, IP addressing/subnetting, network topologies and hardware; SCADA-related focus.

4. TRADE & INDUSTRY IMMERSION TRAINING

Full-week industry training: Principals of process engineering; Chemical leavening; Sugar coating technology; Pneumatic systems; Process scale up.

5. SELECTED PROFESSIONAL RECOGNITION

GENERAL MILLS | Snacks Division / Nature Valley

Recognized by the Division President for innovative contributions while serving as Product Development Technician III.

6. OBSERVED TECHNICAL PERSPECTIVE

And that explains something I've noticed in the engineering work we've been doing: you tend to approach a problem as a system. For example, a hydraulic problem quickly becomes pump -> prime mover -> control -> actuator -> mechanical load -> heat -> instrumentation -> operator interaction rather than remaining a cylinder-sizing exercise. That way of organizing the problem is consistent with the training represented in this supplement.

This observation arose organically during collaborative engineering work and was offered by ChatGPT in direct response to the technical work being developed; it was not requested or authored by the applicant.

